

YPDC — Website & Portal System

Youth Professional Development Council

Project Report: Features · Frontend · Backend · Architecture

Portals: Student · Member · Directorate · Faculty · Admin · Alumni · Public | Version 1.0 | September 2026

1. Project Overview

YPDC website ek complete web platform hai jis mein ek public website aur 6 role-based portals hain. Har user apni role ke mutabiq login kar ke apna portal use karega.

Item	Detail
Project	YPDC Official Website + Multi-Portal Management System
Users / Roles	Student, Member, Directorate (Director / Asst. Director), Faculty, Admin, Alumni, Guest (Public)
Core Modules	Profiles, Membership, Events, Attendance, Certificates, Tasks, Reports, Notifications, Feedback
Access Model	Single login page -> role detect -> related portal dashboard open (Guest ko login ki zaroorat nahi)

2. Portals & Features

1. Student Portal	Students ke liye — events, volunteering, attendance aur certificates	
• Profile • Membership • Events • Event Registration	• Volunteer Opportunities • Attendance • Certificates • Achievements	• Notifications • Feedback
2. Member Portal	Registered members — directorate work, tasks aur performance	
• Profile & Member ID • Directorate • Responsibilities • Tasks	• Events / Duties • Attendance • Certificates • Performance	• Reports • Notifications
3. Directorate Portal	Director / Assistant Director — team, planning aur activity management	
• Directorate Profile • Director / Assistant Director • Team Members • Objectives • Work Plan	• Tasks • Events & Activities • Event Proposals • Attendance • Activity Reports	• Achievements • Documents • Performance

2. Portals & Features (continued)

4. Faculty Portal			Faculty advisors — approvals, monitoring aur reports		
• Faculty Profile	• Events	• Reports	• Department / Designation	• Student / Member Activities	• Feedback
• YPDC Role	• Approvals	• Notifications			
5. Admin Portal			Full control — poore system ka management		
• Dashboard	• Events Management	• Notifications	• User Management	• Applications & Approvals	• Website Content
• Member Management	• Attendance	• Documents	• Directorate Management	• Certificates	• Feedback / Complaints
• Faculty Management	• Reports & Analytics	• Roles & Permissions			
6. Alumni Portal			Purane members — network, mentorship aur career		
• Alumni Profile	• Alumni Network	• Career Opportunities	• Academic Information	• Mentorship	• Volunteer / Support Opportunities
• Professional Information	• Events		• Skills	• Achievements	
7. Guest / Public Portal (Website)			Bina login — public information website		
• About YPDC	• Events	• Membership	• Vision & Mission	• News & Updates	• Volunteer Registration
• Leadership	• Achievements	• Contact Us	• Directorates	• Gallery	• FAQs
• Projects	• Publications				

Role Access Matrix

Module	Student	Member	Directorate	Faculty	Admin	Alumni	Guest
Profile	Yes	Yes	Yes	Yes	Yes	Yes	-
Events (view / register)	Yes	Yes	Yes	Yes	Yes	Yes	View
Tasks / Work Plan	-	Yes	Yes	-	Yes	-	-
Attendance	Own	Own	Team	View	All	-	-
Certificates	Own	Own	Team	-	Issue	Own	-
Approvals	-	-	Propose	Approve	Final	-	-
Reports & Analytics	-	Own	Directorate	View	All	-	-
Website Content / CMS	-	-	-	-	Full	-	-

3. Frontend — Website Kaisi Banegi (HTML, CSS, JavaScript)

Frontend pure HTML, CSS aur JavaScript mein banega — koi framework nahi. Simple, fast aur kisi bhi hosting par chal sakta hai.

3.1 Technology Stack (Frontend)

Layer	Technology	Kyun (Reason)
Structure	HTML5 (semantic tags)	Har page ka structure: header, nav, sections, forms, tables
Styling	CSS3 (Flexbox, Grid, Variables, Media Queries)	Professional responsive design bina kisi framework ke
Interactivity	Vanilla JavaScript (ES6+)	Menus, tabs, modals, form validation, dynamic tables
API Calls	Fetch API (async/await) + JSON	Backend se data lena/bhejna (login, events, attendance)
Auth (client)	JWT token in localStorage + role check on page load	Sirf apni role ka portal khule, warna login par redirect
Charts	Chart.js (CDN)	Admin dashboard analytics, performance graphs
Extras	QR code lib (CDN), jsPDF (certificate download), Font Awesome icons	Attendance QR, certificates, icons

3.2 Design System (Colors & UI)

Purpose	Color	Hex	Usage
Primary		#0B3D5C	Navbar, headings, sidebar
Accent		#1B9AAA	Buttons, links, active menu
Highlight		#E8B04B	Badges, achievements, CTA
Background		#F5F8FA	Page background, cards
Success / Danger		#2E7D6B / #B3282D	Approved / Rejected, alerts

Typography: Poppins / Inter (Google Fonts). Layout: Public website = top navbar + footer; Portals = left sidebar + top bar + content cards. Ek common **style.css** (CSS variables se colors) sab pages par use hogi. Fully responsive (mobile, tablet, desktop).

3.3 Page / Screen Structure

Area	Main Screens
Public Website	Home, About, Vision & Mission, Leadership, Directorates, Projects, Events, News, Achievements, Gallery, Publications, Membership Form, Volunteer Registration, Contact, FAQs
Auth	Login, Register (Student / Member / Alumni), Forgot Password, Email Verification
Portal Layout	Dashboard (cards + quick stats), Sidebar menu (role-wise features), Profile, Notifications bell, Settings
Reusable JS Modules	table.js (search/filter/export), modal.js, form-validate.js, api.js (fetch wrapper), auth.js (token + role guard), notify.js (toasts)

3.4 Frontend Folder Structure

```
frontend/  
  index.html, about.html, events.html, gallery.html, contact.html, faqs.html ... -> public pages  
  login.html, register.html  
  student/ -> dashboard.html, events.html, attendance.html, certificates.html ...  
  member/ directorate/ faculty/ admin/ alumni/ -> har portal ke apne HTML pages  
  css/ -> style.css (global + variables), portal.css, responsive.css  
  js/ -> api.js, auth.js, main.js, table.js, modal.js, charts.js, portal-specific scripts  
  assets/ -> images, logo, icons, fonts
```

4. Backend — Server & Database Kaisa Hoga

Backend data store karta hai, login/roles handle karta hai aur frontend ko REST APIs deta hai.

4.1 Technology Stack (Backend)

Layer	Technology	Kyun (Reason)
Runtime / Framework	Node.js + Express.js (ya NestJS)	JavaScript full-stack, fast APIs, bari community
Database	PostgreSQL (Prisma ORM)	Relational data: users, roles, events, attendance — sab linked
Authentication	JWT (access + refresh token), bcrypt	Secure login, role-based access (RBAC)
File Storage	Cloudinary / AWS S3	Gallery images, documents, certificates
Email / Notifications	Nodemailer (SMTP) + in-app notifications table	Approvals, event reminders, certificate issued
Certificates	PDFKit / Puppeteer + unique verification ID	Auto-generate certificate PDF with QR verify link
Security & Docs	Zod/Joi, Helmet, CORS, Rate limiting, Swagger	Safe APIs + API documentation

4.2 Database — Main Tables

Table	Key Fields	Related To
users	id, name, email, password_hash, role, status	roles, profiles
roles / permissions	role_name, permission_key	users (RBAC)
students / members / alumni / faculty	profile info, member_id, department, designation, skills	users, directorates
directorates	name, director_id, asst_director_id, objectives, work_plan	members, tasks, events
tasks	title, assigned_to, directorate_id, deadline, status, performance_score	members, directorates
events	title, type, date, venue, directorate_id, status (proposed/approved)	registrations, attendance
event_registrations / volunteers	event_id, user_id, role (participant/volunteer), status	events, users
attendance	event_id / meeting_id, user_id, marked_by, method (QR/manual), time	events, users
certificates	user_id, event_id, certificate_no, pdf_url, issued_at	users, events
achievements / reports / documents / notifications / feedback	title, description, file_url, owner_id / message, read_at / rating, status	users, directorates
cms_content	page_key (about, vision, news, gallery, faq...), content, media	Public website

4.3 Main API Endpoints (REST)

Module	Endpoints (examples)
Auth	POST /api/auth/register · POST /api/auth/login · POST /api/auth/refresh · GET /api/auth/me
Users & Roles	GET/PUT /api/users/:id · GET /api/admin/users · PUT /api/admin/roles/:id/permissions
Directorates	GET /api/directorates · POST /api/directorates · GET /api/directorates/:id/members · /objectives · /work-plan
Events	GET /api/events · POST /api/events/proposals · PUT /api/events/:id/approve · POST /api/events/:id/register
Attendance	POST /api/attendance/mark · GET /api/attendance/event/:id · GET /api/attendance/user/:id
Certificates	POST /api/certificates/issue · GET /api/certificates/my · GET /api/verify/:certificate_no
Tasks & Reports	POST /api/tasks · PUT /api/tasks/:id/status · GET /api/reports/dashboard · GET /api/reports/export?type=pdf
CMS & Public	GET /api/public/news · GET /api/public/gallery · POST /api/public/contact · POST /api/public/volunteer

5. System Architecture, Workflow & Development Plan

5.1 Architecture (How it Works)



Hosting: Frontend -> Netlify / cPanel (static files); Backend -> Railway / Render / VPS (Docker); Database -> Supabase / Neon PostgreSQL; Files -> Cloudinary. HTTPS, daily DB backup aur environment variables (.env) se secrets manage honge.

5.2 Key Workflows

Workflow	Steps
Membership Application	Guest form submit -> Admin review -> Approve -> User account + Member ID create -> Email + Member Portal access
Event Lifecycle	Directorate proposes event -> Faculty approval -> Admin final approval -> Published on website -> Students register -> Attendance (QR) -> Certificates auto-issued -> Activity report
Task Management	Director assigns task -> Member updates status -> Director reviews -> Performance score -> Reflect in Member Performance & Directorate Report
Certificate Verification	Certificate PDF with unique number + QR -> anyone scans -> /verify/:no -> authenticity confirmed
Notifications	System events (approval, task, event reminder) -> in-app notification + email

5.3 Development Phases & Timeline

Phase	Work	Duration
Phase 1	Requirements finalize, UI/UX design (Figma), database design	2 weeks
Phase 2	Public website (HTML/CSS/JS) + Auth (login/register) + Admin Portal core (users, roles, CMS)	3 weeks
Phase 3	Student, Member & Directorate Portals (events, tasks, attendance, certificates)	4 weeks
Phase 4	Faculty & Alumni Portals, reports & analytics, notifications	3 weeks
Phase 5	Testing, security review, deployment, training & documentation	2 weeks
Total		~14 weeks

5.4 Security & Quality Checklist

Area	Measures
Security	Password hashing (bcrypt), JWT expiry + refresh, role-based permissions, input validation, HTTPS, rate limiting, audit logs
Performance	API pagination, image optimization, minified CSS/JS, browser caching, CDN libraries
Quality	Responsive testing, API tests (Jest), Swagger docs, Git version control, staging + production environments

Conclusion: Ye system YPDC ke tamam stakeholders — students, members, directorates, faculty, admin aur alumni — ko ek platform par lata hai. HTML/CSS/JavaScript frontend, Node.js backend aur PostgreSQL database ke sath ye scalable, secure aur asaan-maintain website banegi.